

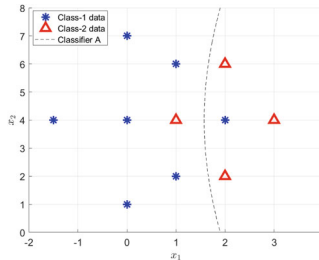
Based on the aggregated percentage populations obtained from thresholding two distributions, the Receiver Operating Characteristic (ROC) curve shows certain resemblance to the Lorenz curve which is based on a single distribution but using two attributes, namely, percentage of population and percentage of wealth or total income. The Lorenz curve is a popular tool for statistical distribution analysis in economics and econometrics (Kleiber, 2008) while the ROC curve is commonly used in analysis related to signal detection which can be dated back to World War II where it was used for analysis of radar signals (Flach, 2010).

The ROC soon found its use in psychology, medicine, biomedical informatics, radiology, biometrics, and other areas related to machine learning and data mining research after World War II. In psychology, the ROC was used to account for perceptual detection of stimuli (Swets, 1996). In clinical medicine and biomedical informatics, the ROC curve was used for evaluating classification and prediction models for decision support, diagnosis, and prognosis (Zweig & Campbell, 1993; Lasko et al., 2005; Zou et al., 2007) of a laboratory test. In radiology, the ROC curve was used as a measure of the accuracy of a diagnostic test (Obuchowski, 2003). In biometrics, the system performance at all operating threshold values was assessed using the ROC curve (Jain et al., 2002; 2004; 2011).

In continuous observations, the AUC can be expressed as the probability that the score of the positive class observation is greater than that of the negative class observation (Bamber, 1975). The AUC can also be considered a normalized form of the Wilcoxon-Mann-Whitney sum of rank test (see Chap. 9) (Mason & Graham, 2002). In multibiometrics, the area under the ROC curve (AUC) was adopted as an optimizing objective for fusion classifier design (Toh et al., 2008b).

### 4.6 Exercises

- 4.1 What is the difference between a regression loss and a classification loss?
- 4.2 What is the difference between a loss function and an error function?
- 4.3 What is the difference between a loss function and a cost function?
- 4.4 What are the common loss functions for regression and that for classification?
- 4.5 The plot below shows the decision boundary of Classifier A. Write down the confusion matrix for Classifier A below.



4.6 Consider the experimental observations of the training error rate (Tr) and the validation error rate (Va) for a machine learning algorithm for each setting of the parameter  $\theta$ . Choose a parameter (P) based on Tr and Va.

| $\theta$ | Tr   | Va   |
|----------|------|------|
| 0.1      | 0.08 | 0.23 |
| 0.2      | 0.33 | 0.38 |
| 0.3      | 0.21 | 0.18 |
| 0.4      | 0.13 | 0.26 |
| 0.5      | 0.16 | 0.18 |

- Which value of parameter  $\theta$  will you choose based on the above observation?
- 4.7 Consider the binary classification problem, which you are dealing with, has highly imbalanced classes. The majority class has 99 hundred samples and the minority class has 1 hundred samples. Which of the following metric(s) would you choose for assessing the classification performance? (select all relevant metric(s))
- (a) Accuracy

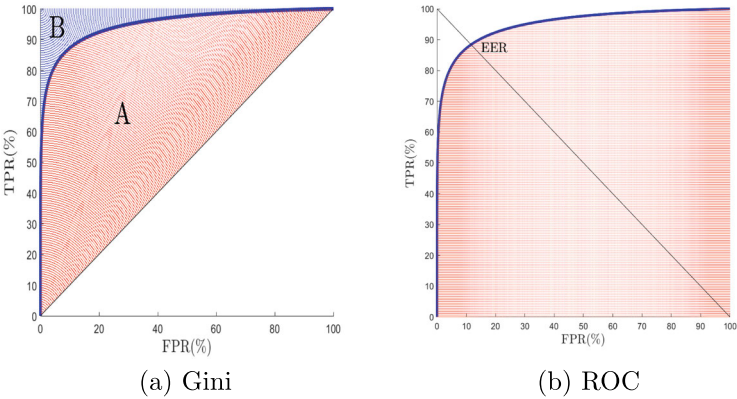
(b) Cost-sensitive accuracy

(c) Precision and recall

(d) Type-I and type-II errors

(d) None of these

4.8 According to the plots of Figs. 4.8b and 4.10 (reproduced below), what is the relationship between the Gini coefficient and the Area Under the ROC (AUC)?



4.9 Given the following confusion matrix for a binary classifier:

|                     | Class(+1) | Class(−1) |
|---------------------|-----------|-----------|
| Predicted Class(+1) | 990       | 20        |
| Predicted Class(−1) | 10        | 380       |

Let  $n^+$  and  $n^-$  denote the numbers of Class(+1) and Class(−1) samples, respectively. Calculate the

- (a) True positive rate
  - (b) True negative rate
  - (c) Accuracy at  $s = n^-/n^+$  (this is the most widely adopted accuracy)
  - (d) Accuracy at  $s = 1$
  - (e) Precision at  $s = n^-/n^+$
- 4.10 The Misclassification Error (MCE) is the count of samples classified incorrectly. Suppose the binary targets are labeled as  $\{-1, +1\}$  and  $g(\mathbf{x})$  denotes the predictor output. Express the MCE in terms of the margin  $y_i g(\mathbf{x}_i)$ .